

CJ Geier

Dr. Wang

Seminar In Data Analytics

October 3rd, 2026

Evaluating the Effect of Play-Action Passing on Defensive Disruption

Football has changed a lot over time, with modern offenses relying heavily on play-action to increase their efficiency. While it's widely accepted that play-action works (Sharp, 2019, 5), there is a gap in understanding exactly how it physically disrupts the defense, the "why". The main goal of this inferential study is to explore that gap by looking at the specific cause-and-effect relationship. I want to see if executing play-action directly causes physical hesitation in the front seven, and if that hesitation ultimately reduces defensive pressure and impacts defensive movement enough to increase offensive efficiency, measured through EPA. Because this project relies on observational tracking data rather than a controlled experiment, I can't definitively claim true causality. Because offensive play-calling is not random, coaches will choose their plays based on confounding variables such as game script, defensive personnel, or pre-snap reads. Even with a large sample size, the effects of these confounders are not totally removed. Instead of guaranteeing true cause and effect, I will state a strong statistical association by controlling for observable game situations such as the down and distance. If this method of matching these game situations does not give me enough overlap between the play types, my backup design strategy will be to use nonparametric inference, specifically the Mann-Whitney-Wilcoxon test. Because football metrics like EPA do not follow a perfectly normal distribution, this test bypasses the strict statistical assumptions. However, changing the test does not automatically solve a lack of comparable plays, and it is not a general median test

without additional conditions. There is a risk of non-objective results when using the Mann-Whitney-Wilcoxon test, so while I will still use it, I will not be able to rely too heavily on it (Thrush, 2021, 6).

I will use secondary observational data collected by the NFL for the NFL Big Data Bowl and made available on Kaggle (National Football League, 2024, 3). This dataset is open-access for public research, giving me permission to work with it to analyze and share any findings without ethical or privacy concerns. The data comes in two parts: play-by-play context, where the unit of analysis is the play itself, and tracking data, which uses the individual player as the unit of analysis. To conduct my research, I will aggregate these sources by joining the high-frequency tracking metrics with the play-by-play situations through Play-ID, a unique categorical identifier. To ensure that my work is reproducible, the raw datasets will be kept in a dedicated folder while I perform data aggregation and cleaning via R scripts in a separate code folder. All data aggregation, visualization, and statistical modeling will be done with the R programming environment (R Core Team, 2024, 4), and will primarily use the tidyverse packages for data manipulation (Wickham, 2019, 7). I will also keep all my documentation clean in GitHub to maintain clear version control and allow anyone reviewing the project to track my workflow and replicate my results.

When preparing this data in R, I will be working with data that captures both the play context and the physical player movement. The main independent variable I will test is a binary boolean (true or false) for whether the offense used play-action. My dependent variables will measure the effects on both the defense and the offense. One of the most important dependent variables will be all of the raw x and y tracking coordinates, which are continuous numeric data types. However, this is not the only data I will be using to measure defensive disruption. I will

also be using a few of the metrics already provided in the dataset. To measure defensive hesitation, I will use existing continuous numeric dependent variables like pass rush get-off and time to pressure. I will also evaluate pass rush win rate as a continuous aggregated percentage dependent variable across multiple plays. These will all help show how the defenses are disrupted when play-action is called through their movement patterns and efficiency at rushing the passer (Eager & Seth, 2023, 1). EPA, which measures a play's value by tracking the change in expected points before and after the snap, will also be used as a continuous numeric dependent variable to measure offensive efficiency when using play-action, as it is the best way to objectively measure offenses' success on plays. I will use only first, second, and third downs, with the maximum distance to a first down being 10 yards. This will ensure that running the ball is a realistic option for a play call, therefore making play-action a realistic option as well. I will make sure to exclude fourth downs, quarterback spikes, trick plays, and garbage time situations where the score is a blowout, because these scenarios can completely change how a defense normally plays and would skew my results (McGarrity & Linnen, 2010, 2). Although the inclusion criteria seem to be strict, the original dataset spans an entire NFL season, so even after filtering, there will still be thousands of qualifying plays to analyze. While this large sample size should provide high statistical precision and power across my models, it does not remove the effect of confounding variables. Finally, to address the issues that arise from missing data with the RFID tracking chips, I will filter out any plays where the sensor data jumps unrealistically to make sure all of the provided metrics are accurate.

For the statistical design, my primary method will be situational matching to better isolate the effect of play-action. To be clear, situational matching ensures I am comparing similar play situations by pairing a play-action pass with a standard dropback that occurred under extremely

similar game conditions. I will tightly control for the play context by matching plays based on their specific down and distance, field position, and the current score differential. While there is a risk that controlling the situations too much could inadvertently introduce new confounding issues, I am not overly concerned about this. This is because the variables I am matching on are strictly pre-snap conditions, meaning that they are more like environmental controls rather than controlling post-snap outcomes. This means it should help the results by safely isolating the play-call without creating any new bias. This will make sure the only actual difference between the two matched plays is whether play-action was called. Once the plays are paired up, I will use specific regression models to analyze the results rather than just looking at basic averages. For my continuous dependent variables like pass rush get-off, time to pressure, EPA, and the aggregated pass rush win rate, I will run multiple linear regressions. The mathematical formula for this method is expressed as:

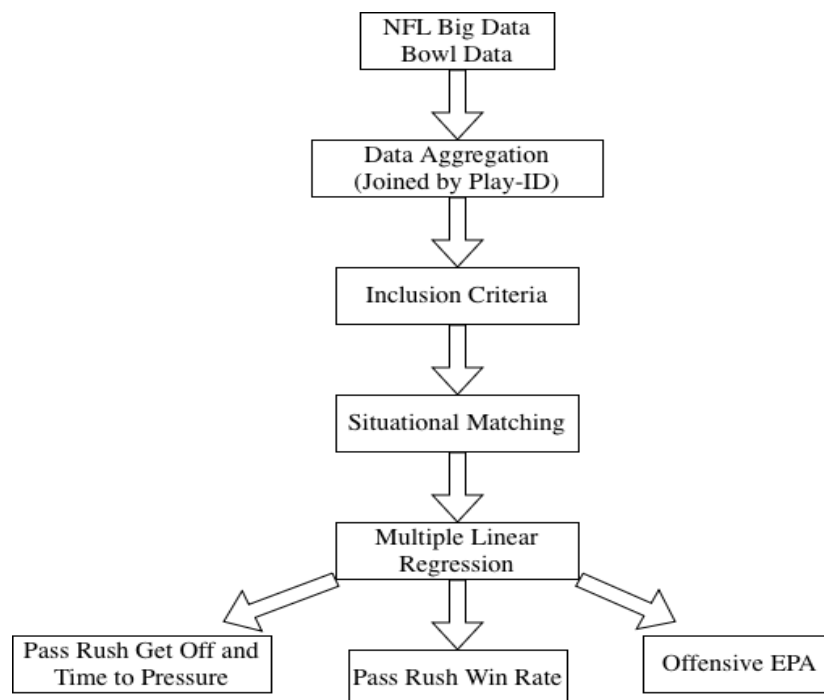
$$Y = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + \dots + \epsilon,$$

where Y represents the continuous outcome metric, β_0 is the intercept, β_1 is the coefficient for the play-action boolean (X_1), and the additional βX terms represent any of the situational control variables mentioned above. In all of these models, the metric will be the outcome variable, and the play-action boolean will be the main explanatory variable, allowing for me to very accurately estimate the weight that play-action has on the outcome of the play. As I mentioned earlier, if I cannot find enough perfect matches across the dataset to run these regressions safely, my backup design will be to use the Mann-Whitney-Wilcoxon test to compare these dependent variables.

Keeping in mind that it only functions as a true median test if the distributions are identically shaped (Thrush, 2021, 6).

To make sure my analysis and results are structurally sound, I will run a series of specific diagnostic validation checks on the outputs. For the multiple linear regressions analyzing continuous variables like EPA and time to pressure, I will analyze residual plots and use the Breusch-Pagan test to evaluate homoscedasticity, and I will run a Shapiro-Wilk test to check that the errors are normally distributed so the model's estimates are not biased. Finally, I will test for multicollinearity using the Variance Inflation Factor (VIF) to check that my control variables are not highly correlated with each other. This will help verify that the statistical weight assigned to the play-action variable is not being inflated by situational constraints like down and distance or any other factors.

Figure 1: Conceptual Model of Methods



References

1. Eager, E., & Seth, T. (2023). The mechanics and efficiency of play-action passing in modern NFL offenses. SumerSports.
2. McGarrity, J. P., & Linnen, B. (2010). Pass or run: An empirical test of the matching pennies game in the National Football League. *Southern Economic Journal*, 77(1), 102–110.
3. National Football League. (2024). NFL Big Data Bowl 2024: Player tracking data and analytics. NFL Football Operations.
4. R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing.
5. Sharp, W. (2019). Play-action passing efficiency and defensive run commitment in professional football. Sharp Football Analysis.
6. Thrush, K. (2021). Evaluating the limitations of expected points added in sub-second play-calling deception. *Journal of Quantitative Analysis in Sports*.
7. Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemond, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., . . . Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686.